



Meningitis

Meningitis is the inflammation of the membranes called the meninges, which surround and protect the brain and spinal cord. It may develop as a result of infection, but it can also occur due to certain non-infectious causes. Bacteria, viruses, fungi, and parasites can cause meningitis. Bacterial meningitis, in particular, is a serious disease that can progress rapidly and be life-threatening. Therefore, early medical evaluation is important when meningitis is suspected.

Meningitis can occur in all regions of the world; however, its incidence varies considerably between regions. One of the regions where bacterial meningitis is most prevalent is the area in sub-Saharan Africa extending from Senegal to Ethiopia, known as the “meningitis belt.” According to sources, approximately 2.5 million cases of bacterial meningitis and 300,000 deaths were reported worldwide in 2024. Viral meningitis is more common than bacterial meningitis and is generally milder in its course.

The symptoms of meningitis may vary depending on the underlying cause of the disease and the rate at which it progresses. The most common symptoms include fever, severe headache, neck stiffness, nausea and vomiting, sensitivity to light, and changes in consciousness or mental status. Seizures, weakness, and skin rash may also occur in some patients.

Bacterial meningitis may progressively worsen over several days in some individuals, whereas in some cases, symptoms may develop much more rapidly. Particularly in meningococcal infection, the disease may become severe within a few hours and progress to sepsis and circulatory dysfunction. In infants, the symptoms may be different and may include difficulty feeding, excessive sleepiness, irritability, persistent crying, and bulging of the fontanelle.

The causative agents of meningitis include bacteria, viruses, fungi, and parasites. Important causes of bacterial meningitis include *Streptococcus pneumoniae*, *Neisseria meningitidis*, *Haemophilus influenzae*, and group B streptococci. In viral meningitis, non-polio enteroviruses are particularly prominent.

The risk increases with conditions such as young age, weakened immune system, inadequate vaccination, and living in crowded environments. In addition, head trauma, brain surgery, cerebrospinal fluid leakage, absence or damage of the spleen, and certain chronic diseases are also associated with an increased risk. The risk is particularly elevated in individuals with cerebrospinal fluid leakage, those who have had their spleen removed or whose spleen is damaged, and those who have experienced head trauma.

The causative agents of meningitis may reach the central nervous system through the bloodstream, by using neural pathways, or directly from nearby infected areas. When bacteria reach the areas containing

cerebrospinal fluid, they multiply and initiate a strong inflammatory response. This process may lead to disruption of the blood-brain barrier, cerebral edema, and ischemia. The resulting inflammation and structural changes cause damage to nerve cells and contribute to the development of serious consequences of the disease.

In the diagnosis of meningitis, in addition to the medical history and physical examination, one of the most important investigations is the evaluation of cerebrospinal fluid (CSF) obtained through lumbar puncture. Cell count, glucose, protein, Gram staining, and culture studies may be performed on the CSF. PCR and other rapid molecular tests may also be used to identify the causative agent. Blood tests and blood cultures also support the diagnosis.

If bacterial meningitis is suspected, treatment should not be delayed because of diagnostic procedures. In some patients, computed tomography may be required before lumbar puncture.

Meningitis is a condition that requires urgent medical intervention. When bacterial meningitis is suspected, antibiotic treatment should be initiated as soon as possible. In some cases, corticosteroids such as dexamethasone may be administered together with antibiotics.

In viral meningitis, most mild cases may resolve with rest, adequate fluid intake, and control of fever or pain without



the need for specific treatment. Antiviral treatment may be used in cases caused by herpes viruses or influenza. Fungal meningitis is generally treated with long-term, high-dose intravenous antifungal medications. Treatment is determined according to the underlying cause of meningitis.

The outcomes following meningitis vary depending on the causative agent, the patient's age, and immune status. In particular, bacterial meningitis can result in death and may leave permanent effects in survivors, including hearing loss, seizures, learning and cognitive problems, visual or motor impairments, and memory problems. Cerebral edema, hydrocephalus, and various neurological disorders may also develop. Although viral meningitis generally has a more favorable course, serious neurological consequences may occur in some cases.

Vaccination is one of the most important methods of preventing meningitis. Vaccines against Hib, pneumococcus, and meningococcus are available. Particularly in the African meningitis belt, the introduction of the five-component meningococcal vaccine providing protection against serogroups A, C, W, Y, and X represents an important development in reducing outbreaks.

In conclusion, meningitis is a disease characterized by inflammation of the membranes surrounding the brain and spinal cord and, particularly in its bacterial form, may lead to serious consequences. It may manifest with symptoms such as fever, severe headache, neck stiffness, and changes in consciousness. While examination of cerebrospinal fluid is particularly important in diagnosis, treatment is provided according to the causative agent using antibiotics, antivirals, or antifungal medications. Early diagnosis and appropriate treatment are of great importance in reducing the risk of death and permanent neurological damage. Vaccination is one of the most effective methods of preventing meningitis.

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